

- Chapter 1 of Python Machine Learning by Example

Data

- Lots of data available
- Can be in variety of formats
- How do you parse ?
- Is the data clean, how much noise ?
- Data for ML algorithms divided into training and testing sets
- Training set used to derive patterns to build **ML model**
- Testing set used to analyze performance of model

Generalization & Goodness of Fit

- **Generalization** refers to how well the concepts learned by a ML model generalizes to specific examples or data not yet seen by the model.
 - E.g., Image of cat correctly identified
 - Would identification be possible even if the cat is moved to a corner or if turned upside down ?
 - Humans generalize with ease but ML algorithms struggle
- **Goodness of fit** describes how well a model fits for a set of observations.
 - Overfitting and Underfitting

Overfitting

- Lets consider practice questions of an exam.
Given 5 practice questions:
 - If there are two potatoes and one tomato in a question, answer happens to be A
 - If there is one potato and three tomatoes in a question, answer happens to be B
 - Using this pattern, you answer your exam
 - Or you memorize each of the 5 questions and answer
 - You may score high in practice
 - However your score low in actual exam !!!

Overfitting

- Phenomenon of extracting too much information from training sets or memorization can cause overfitting
 - Makes ML model work well with training data called **low bias**
 - Bias refers to error due to incorrect assumptions in learning algorithm
 - However, does not generalize well or derive patterns, performs poorly on test datasets called **high variance**
 - Variance measures error due to small fluctuations in training set

Overfitting

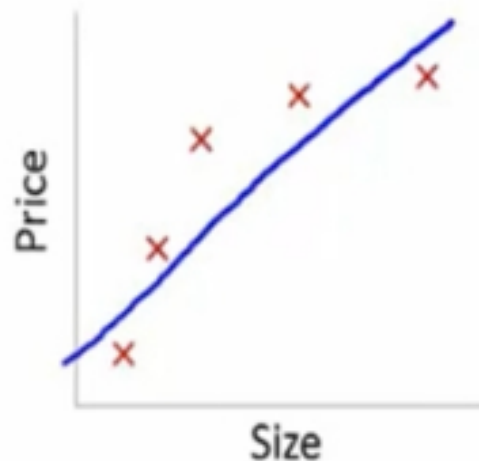
- In summary:
 - A well functioning ML algorithm separates signal from noise
 - If algorithm is too complex or too flexible (e.g., uses too many features), it can end up memorizing the noise instead of finding the signal
 - Overfit model can therefore end up making predictions based on the noise
- Occurs when we generalize using too few examples or build excessively complex model to fit every training sample (e.g. memorization)

Underfitting

- Model is underfit if it does not perform well on training sets and will not do so on test sets
- Occurs when we are not using enough data to train or if we try to fit wrong model to the data
 - E.g., if you do not read enough material for exam or if you prepare wrong syllabus
- Called **high bias** in ML although **variance is low** [i.e. consistent but in a bad way]
- May need to increase number of features since it expands the hypothesis space.

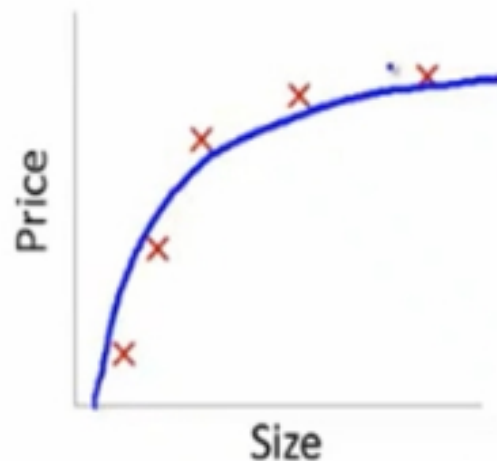
Goodness of fit

- For same data:



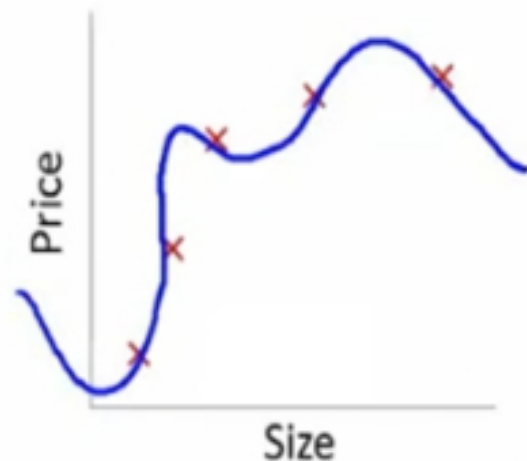
$$\theta_0 + \theta_1 x$$

High bias
(underfit)



$$\theta_0 + \theta_1 x + \theta_2 x^2$$

“Just right”



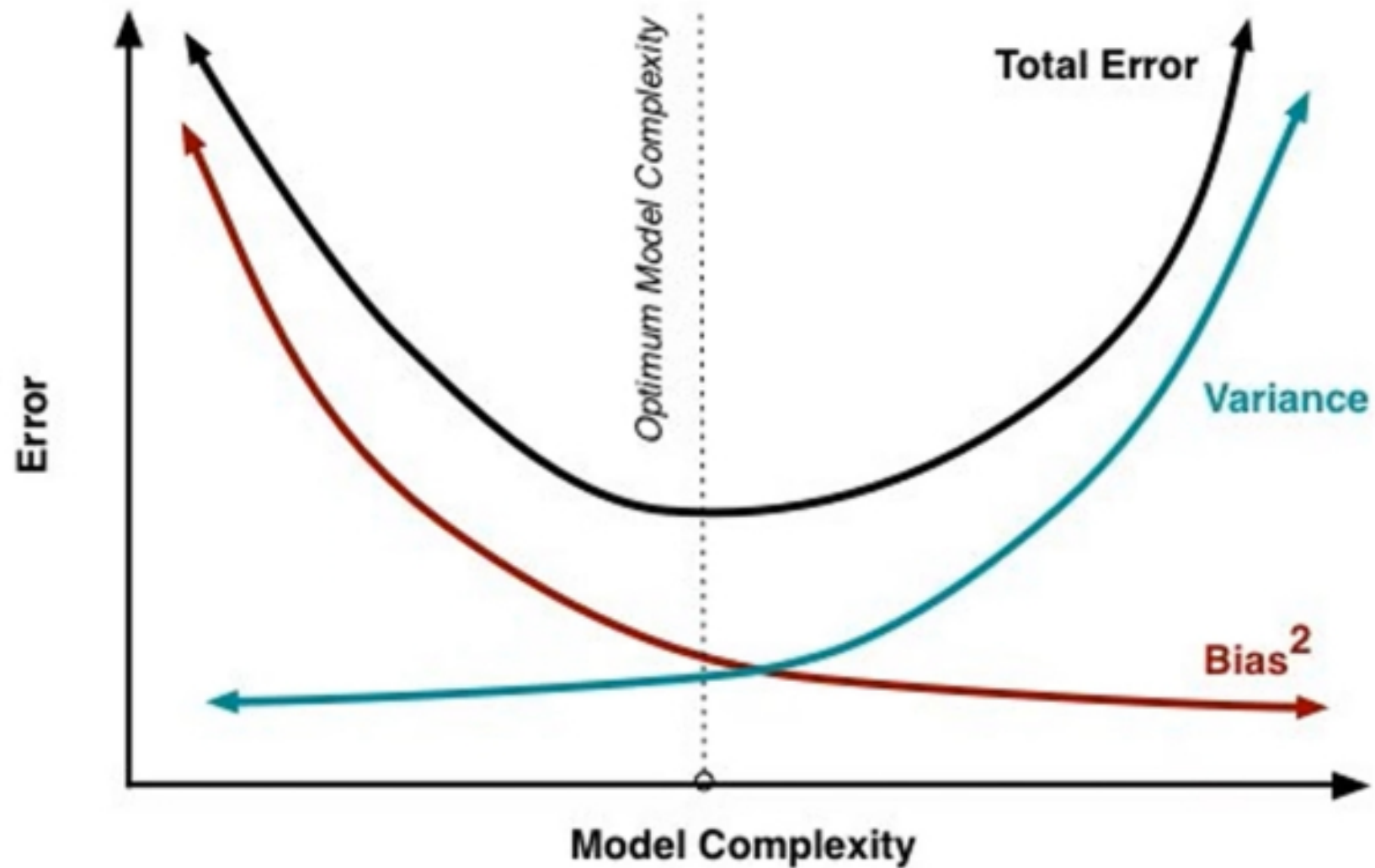
$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

High variance
(overfit)

Bias-Variance Tradeoff

- If the model is too simple and has very few parameters then it may have high bias and low variance
- If the model has large number of parameters it may have high variance and low bias
- We need to find a right/good **balance** without overfitting or underfitting the data
- As more parameters are added to a model
 - Complexity of the model rises
 - Variance becomes primary concern while bias falls steadily.

Bias-Variance Tradeoff



Aim is to minimize error of learning model

Bias-Variance Tradeoff

- Suppose a training set consists of points $x_1, \dots, x_n$ and real values y_i associated with each point x_i
- We assume there is a function $y = f(x) + \varepsilon$, where the noise ε has zero mean and variance σ^2
- Find $\hat{f}(x)$, that approximates $f(x)$ as well as possible
- To measure how well the approximation was performed, we minimize the mean square error $(y - \hat{f}(x))^2$
- A number of algorithms exist to find $\hat{f}(x)$, that generalizes to points outside of our training set

Bias-Variance Tradeoff

- Variance measures how far a set of (random) numbers are spread out from their average value.
- Measured as expectation of the squared deviation of a random variable from its mean.

$$\text{Var}(X) = E[(x - \mu)^2]$$

$$\text{Var}(X) = E[(x - E[x])^2]$$

$$= E[x^2 - 2xE[x] + E[x]^2]$$

$$= E[x^2] - 2E[x]E[x] + E[x]^2$$

$$= E[x^2] - E[x]^2$$

Bias-Variance Tradeoff

- Turns out expected (mean squared) error of $\hat{f}$ on an unseen sample in general can be decomposed as:

where,

$$E \left[\left(y - \hat{f}(x) \right)^2 \right] = (\text{Bias}[\hat{f}(x)])^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

$$\begin{aligned} \text{Bias}(\hat{f}(x)) &= E[\hat{f}(x) - f(x)] \\ &= E[\hat{f}(x)] - E[f(x)] = E[\hat{f}(x)] - f(x) \end{aligned}$$

Since f is deterministic, $E[f] = f$

and

$$\text{Var}[\hat{f}(x)] = E[\hat{f}(x)^2] - E[\hat{f}(x)]^2$$

Note that all three terms are positive

Bias-Variance Tradeoff

- Notations:

$$\text{Var}[x] = E[x^2] - (E[x])^2$$

$$E[X^2] = \text{Var}(X) + (E[x])^2$$

Given $y = f + \varepsilon$ and $E[\varepsilon] = 0$, $E[y] = E[f + \varepsilon] = E[f] = f$

Since $\text{Var}[\varepsilon] = \sigma^2$, $\text{Var}[y] = E[(y - E[y])^2] = E[(y - f)^2]$

$$= E[(f + \varepsilon - f)^2] = E[\varepsilon^2] = \text{Var}[\varepsilon] + (E[\varepsilon])^2 = \sigma^2$$